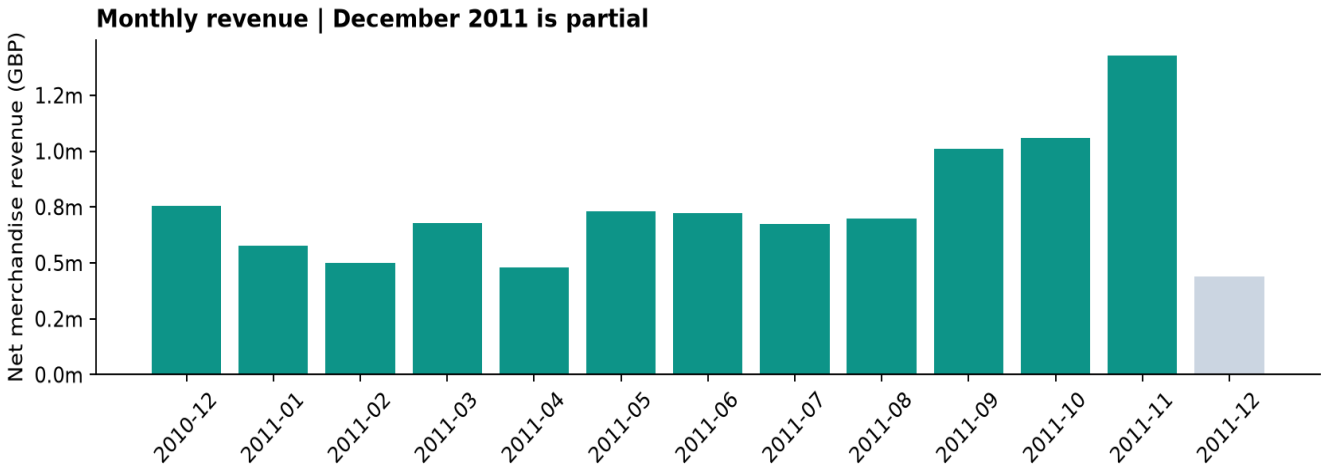
DECISION BRIEF / 01

Prioritize revenue quality, then retention

The analytical merchandise scope contains GBP 10,229,905 in gross sales and GBP 475,249 in cancellation adjustments, leaving GBP 9,754,656 in net merchandise revenue.

Recommended sequence: validate extreme cancellation pairs, investigate high-value product issues, and test retention outreach with identified repeat buyers. Do not turn cancellation value into a claimed saving without a pilot.

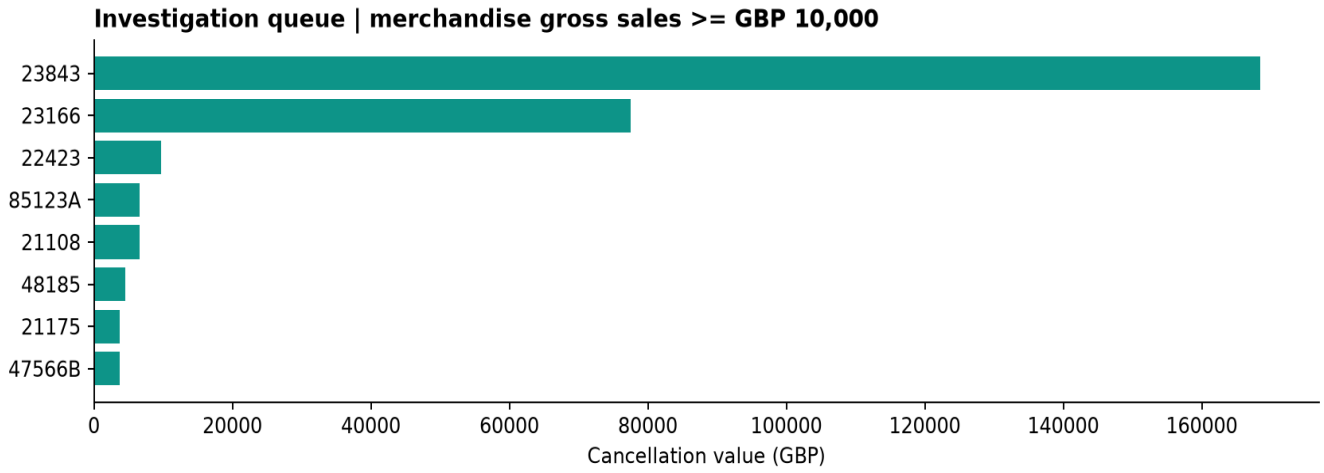


Scope: 541,909 original lines; 5,268 exact duplicates excluded; 19,771 merchandise sales invoices. 14.7% of gross sales lacks a customer ID and is excluded from customer analysis only.

This is revenue analysis, not profit accounting. Costs, discounts, fulfillment status and matched returns are unavailable. December 2011 covers only through December 9 and should not be compared with a full month.

Validate cancellation outliers first

The same-period cancellation-value ratio is 4.65%. Excluding lines with absolute quantity above 10,000, the sensitivity ratio is 2.30%. This difference matters before sizing a recoverable opportunity.



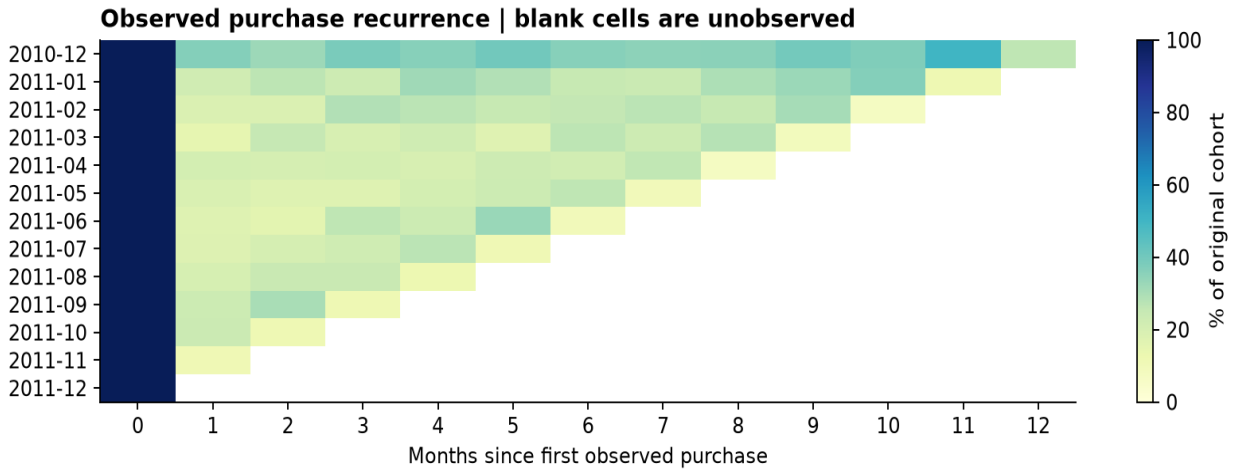
Extreme lines represent 4 analytical rows and GBP 245,653 of cancellations. Large equal-and-opposite transactions may be booking corrections or unusual orders. The source cannot establish their cause.

Action: reconcile invoice history for the largest stock codes before contacting operations. Then inspect product descriptions, order quantities and cancellation timing. The product action list uses a GBP 10,000 gross-sales floor to avoid small-denominator rankings.

A C invoice identifies a cancellation adjustment, not a verified physical return. Cancellations may relate to earlier sales outside the selected window. Some product ratios can exceed 100%; do not interpret them as order-level probabilities.

Protect valuable repeat relationships

Among 4,334 identified buyers, 65.3% purchased on two or more invoices and generated 92.8% of identified gross sales. The top buyer decile contributed 61.4%.



Test a service-led retention pilot with repeat buyers: stock availability notifications or proactive order support. Use an untreated comparison group and measure incremental repeat purchase and net revenue; avoid blanket discounts without cost data.

Cohorts are first observed purchase months, not true acquisition cohorts. The earliest cohort is left-censored and later cohorts have less follow-up. Grey/blank cells are unobserved. Anonymous orders remain in total revenue but never become one artificial customer.

SENSITIVITY / 04

Treat financial upside as a scenario

No observed cost or gross-margin fields exist. The following arithmetic is an illustration of assumptions, not measured profit, ROI, or a forecast.

If 5% of observed cancellation value were preventable: retained revenue = GBP 23,762; at an assumed 35% contribution margin, illustrative contribution = GBP 8,317 before intervention costs.

If 10% of observed cancellation value were preventable: retained revenue = GBP 47,525; at an assumed 35% contribution margin, illustrative contribution = GBP 16,634 before intervention costs.

If 20% of observed cancellation value were preventable: retained revenue = GBP 95,050; at an assumed 35% contribution margin, illustrative contribution = GBP 33,267 before intervention costs.

The dashboard lets reviewers vary cancellation avoidance from 0-30% and assumed contribution margin from 0-80%. Use the outlier comparison from page 2 before selecting a credible addressable base.

Next information to request: original-invoice links, product cost, shipping and handling cost, discount history, reason codes, stock availability and customer contact permissions. Only then build a net-benefit or pricing case.

Investment gate: approve a limited pilot only after establishing a measurable eligible population, a credible baseline, incremental operating cost and a stop rule. Scale if the comparison group shows positive incremental contribution after costs.

Deliver a focused 30-day investigation

Week 1 / Finance + data: reconcile highest-value cancellation pairs and confirm which rows are booking corrections. Deliver an agreed revenue bridge and exclusions register.

Week 2 / Operations: review a sample of high-value product cancellations and map reason codes. Separate availability, order-entry and fulfillment issues before proposing changes.

Week 3 / Customer team: design a controlled outreach test for eligible repeat buyers. Fix success metrics and observation windows before launch.

Week 4 / Sponsor: review reconciliation findings and pilot design. Fund only actions with a clear owner, measurable baseline and plausible incremental benefit.

Validation: 12/12 checks passed, including monetary reconciliation, duplicate accounting, invoice grain, cohort denominators, and country-month additivity. SQL outputs and a data-quality register are committed alongside this brief.

Source: Chen, D. (2015), Online Retail, UCI Machine Learning Repository, DOI 10.24432/C5BW33, CC BY 4.0. This project transforms the data through normalization, duplicate removal, scope classification and aggregation.

Reproducibility: run python src/pipeline.py --download, then python -m unittest discover -s tests -v. Open dashboard/index.html locally. Raw data and customer-level records are not committed.